GeoMatch: photo geolocation as visual retrieval

Mohan Manideep Danda

1 Why retrieval

We predict by **retrieval**: embed query and bank, copy the nearest match's coordinate. Retrieval is not continuous — its output is restricted to the 11,758 training coordinates — but that discretisation is cheap: the geographically nearest training photo is a median 3.29 km away and within 50 km for 99.8% of queries, far below the error we achieve, whereas a geo-cell classifier is bounded by its cell size. The gain is that the encoder never regresses a coordinate; it only learns which photos were taken near each other, which supervises densely from every pair in a batch.

We built the geo-cell alternative first — the same trunk with six hierarchical classification heads and a combinatorial partition decoder. It reached 75.2 km; we dropped it for the retrieval line, which was *worse* then (89.2 km) but had the lower floor and overtook it once self-supervised pretraining was added. Table 1 keeps that non-monotonic ordering rather than hiding it.

2 Model (Fig. 1)

One RegNetY-400MF trunk [1] (`weights=None`) is shared across three views: the full frame at 512^2 and aligned left/right crops at 256^2, which keep detail a single downscaled frame destroys.

Global branch. Multi-scale GeM pooling [2] over stages 2–4, fused across views, through an identity-initialised 384×384 layer, L2-normalised to a descriptor g. **Local branch.** Stage-4 features average-pooled to 4×4 per view (48 tokens), projected $440 \rightarrow 64$, L2-normalised to tokens T.

The shortlist is the cosine top-200 of $\langle g_q, g_b \rangle$, rescored by a symmetric MaxSim late interaction [3]:

$$s(q,b) = 0.2\,\langle g_q, g_b \rangle + 0.8 \cdot \tfrac{1}{2}\left(\overline{\max_j T_q T_b^\top} + \overline{\max_i T_q T_b^\top} \right)$$

The top-1 candidate's coordinate is the prediction.

3 Training

Evaluated on the official 5-fold split: train on four folds, predict the fifth, fixed epoch count, no checkpoint selection.

(1) BYOL pretraining [4] of the trunk, labels unused, 160 epochs. **(2) Country-aware retrieval fine-tuning**, 40 epochs, AdamW (8×10^{-5} trunk, 6×10^{-4} heads), physical batch 32 with 2 gradient-accumulation steps (effective 64), EMA 0.999. Each group is an anchor, two geographic positives and one *same-country* hard negative 60–700 km away, re-mined from the current EMA encoder every epoch. Positives are the 8 most descriptor-similar of the 64 geographically nearest that lie within 50 km; for the 10.7% of anchors with fewer than 8 such neighbours the sampler falls back to the 8 most descriptor-similar of those same 64, *without the 50 km filter*, so "≤50 km" holds for ≈ 89% of anchors, not all. Anchors from the weakest countries are oversampled (DE 3.0×, FR 2.8, PL 2.0, IT/ES 1.7, SE 1.4, GB 1.3). Objective: 0.65 retrieval + 0.2 classification regulariser.

4 Results

pooled 5-fold OOF	runs	med.	mean	<200	<750
Geo-cell classification (dropped)	1	75.2	—	—	—
From-scratch retrieval baseline	1	89.2	618	55.9%	69.4%
+ BYOL + country-aware, 22 ep	1	58.0	490	60.7%	76.2%
+ 40 epochs (shipped recipe)	3	**56.8**±1.1	469	61.4%	77.1%
(non-compliant) 3-model ens.	1	52.0	—	—	—
(non-compliant) 6-model ens.	1	49.7	—	—	—

Table 1: Median/mean in km. Shipped row is mean ± s.d. of three seeds (55.60 / 57.12 / 57.71); **best observed 55.60**, but the submitted model uses the config's default seed, not among the three — no seed was chosen on results. Geo-cell mean/rate columns were not retained. Ensembles are recorded as a real gain rejected on the one-model rule.

Which half did the work? The −31 km row changes two things at once and we cannot cleanly separate them: the available fold-0 runs differ in epoch count as well as in condition, so the differences below confound the two. What we observed on fold 0: baseline (no BYOL, no country-aware, ep 6) 76.1 km; country-aware fine-tuning without BYOL 70.0 km from baseline weights (ep 12) and 71.9 from a joint start (ep 24); with the BYOL trunk **53.6** (ep 22). The runs that include BYOL are the lowest by a wide margin, but no run isolates either change at a matched schedule, so we do not assign a per-component figure. Extending 22 to 40 epochs is within the ±1.1 km seed spread, so we do not claim it as an improvement.

5 Where the error comes from

Splitting each query into *located* (≤50 km), *mis-ranked* (a ≤50 km bank image was in the top-200 but not returned first) and *never retrieved* separates the two failure modes.

	located	mis-rank	not retr.	bank<1 km
Pooled	49.1%	30.3%	20.7%	30.1%
Iceland	82.9%	15.2%	1.9%	57.6%
France	25.2%	32.7%	42.1%	14.5%
Germany	14.2%	40.6%	45.2%	9.1%

Table 2: Computed on a single seed (the 55.60 km run), not averaged over the three of Table 1. Last column: share of queries with *any* bank photo within 1 km. Full 12-country version in the repository README.

Germany (505 km median) and France (570 km) dominate what remains. They are 17% of the evaluation set, and the pooled median over the other ten countries alone is **31.7** km. Ranking is the larger loss pooled, but for

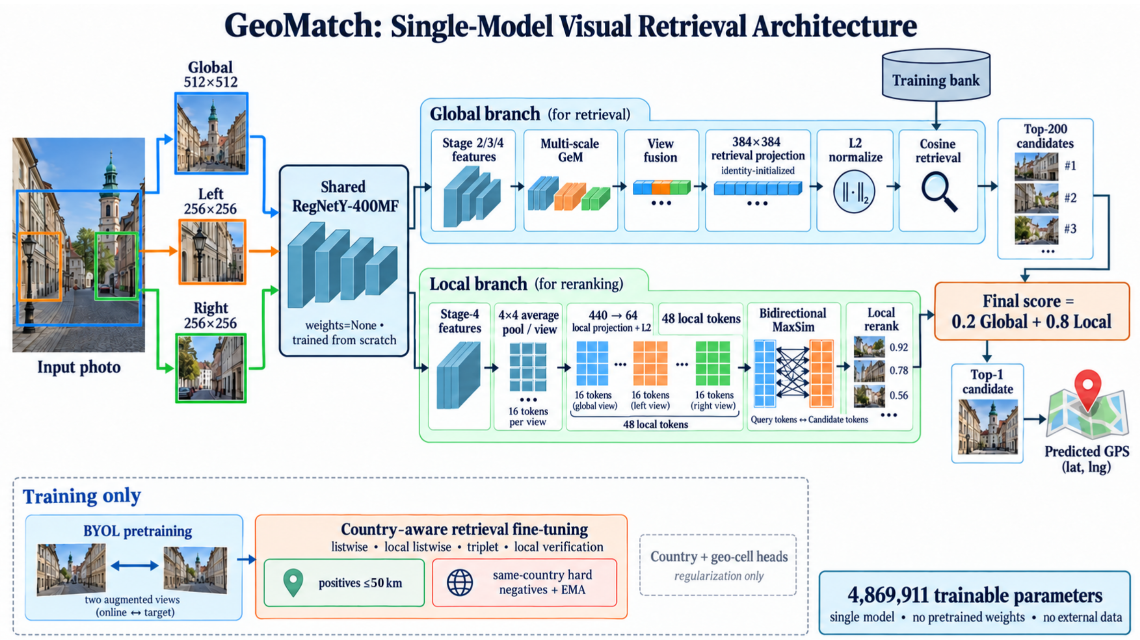

Figure 1: One from-scratch RegNetY-400MF trunk, shared across a global view and two crops, feeds a global-descriptor retrieval branch and a local-token late-interaction reranker. Classification heads regularise training only. 4,869,911 parameters.

these two nearly half of queries retrieve nothing nearby at all — both stages fail, so a better reranker could not have closed it.

Bank coverage tracks performance. Ranking the twelve countries by how often a bank photo lies within 1 km reproduces the performance ordering almost exactly (Spearman −0.89 against median error): Iceland 57.6%, Germany 9.1%. This is an association between the bank's geographic coverage and accuracy; we did not test what drives it, and photos within 1 km need not be views of the same scene. It also does not explain the failure: the nearest German bank photo is a median 8.8 km away against a 505 km German median, so the shortfall is not that no usable neighbour exists.

6 What did not work

Decode-time repairs on frozen descriptors (5-fold evidence). A learned candidate reranker was tried four times (linear, residual, pairwise, and one gated on a held-out inner fold) and degraded the crossfit every time; the inner-fold gate passed only because the probe fold was held out from the reranker but not from the *encoder*. Kernel-density consensus decoding gave 72 km against 57.7. Gating the shortlist on predicted country cost 4 km pooled (country accuracy is 68%, and a wrong gate is unrecoverable). Decoding from the auxiliary geo-cell head instead of retrieval gave DE 490 / FR 523 km against that encoder's retrieval 534 / 551: two decoders reading one descriptor fail equally, which locates the problem upstream of the decoder — though it does not establish what the descriptor is missing.

Encoder variants (fold 0 only — weaker evidence). Adding a random-resized-crop to the global view was worse at every matched epoch (63.7 vs 58.1 km at epoch 32), consistent enough to call a backfire; it discards the horizon and scene layout. Raising the local grid from 4×4 to 8×8 was *not* worse (58.3 vs 60.7 km at epoch 40) — we kept 4×4 as the simpler option, not because the variant failed; extending it to 48 epochs drifted to 61.8 km. Interpret both cautiously: the shipped recipe's own fold-0 median varies by 5.3 km across epochs 20–40, so single-fold gaps of that size are not decisive.

7 Limitations

Germany and France are the dominant failure and we did not solve them. Both stages break there: ∼45% of queries retrieve no nearby image into the top-200, and of those that do, most are then mis-ranked. Swapping the decoder does not help (§6), which points at the shared descriptor, but we never ran the experiment that would say what it is missing — e.g. probing whether it carries any within-country position signal — so we cannot claim the failure is purely representational. Also: the ablation and failed-variant evidence is single-fold and confounded by epoch count, and the figures are rebuilt from cached out-of-fold predictions that are not in the repository, so they cannot be regenerated from a clean clone.

8 Submission

The recipe retrained once on all 11,758 images with no held-out fold, from an existing BYOL trunk, using the config's default seed: 4,869,911 parameters, **67 min of fine-tuning** on one RTX 4000 Ada (excluding BYOL pretraining, ≈2 h 40 min, reused rather than re-run), plus ≈5 min to write `predictions.csv`.

[1] Radosavovic et al. Designing Network Design Spaces. CVPR 2020.
[2] Radenović, Tolias, Chum. Fine-tuning CNN Image Retrieval with No Human Annotation. TPAMI 2019.
[3] Khattab, Zaharia. ColBERT: Late Interaction over BERT. SIGIR 2020.
[4] Grill et al. Bootstrap Your Own Latent. NeurIPS 2020.